

# **Project: Rocketry Training - Aerospace Simulation Workshop Project**

## **PART A ---> THEORETICAL SECTION**

Q1. Explain Finite Element Method (FEM) in your own words.

Finite Element Method (FEM) is a numerical analysis technique used to predict how a structure behaves under different loads and conditions. It divides a complex object into many small elements, making calculations easier. In aerospace engineering, FEM is used to analyze stress, deformation, vibration, and structural safety of aircraft and rocket components.

Q2. What is Von Mises Stress and why is it important?

Von Mises stress is a value used to predict whether a material will yield under complex loading conditions. If the Von Mises stress exceeds the material's yield strength, the component may permanently deform or fail. It helps engineers design safe and reliable structures.

Q3. Explain the role of meshing in FEM simulations.

Meshing divides a model into small finite elements for numerical analysis. A finer mesh provides more accurate results but requires more computation time, while a coarser mesh is faster but less accurate. Mesh independence ensures that results remain consistent even after further mesh refinement.

Q4. Differentiate between CFD and FEM.

| FEM                                        | CFD                                            |
|--------------------------------------------|------------------------------------------------|
| Analyzes structural behavior               | Analyzes fluid flow                            |
| Calculates stress, strain, and deformation | Calculates pressure, velocity, and temperature |
| Used for solid structures                  | Used for liquids and gases                     |
| Common in structural design                | Common in aerodynamics and heat transfer       |

Q5. Why are boundary conditions important in simulations?

Boundary conditions define how a model is supported, loaded, and constrained during simulation. Correct boundary conditions produce realistic results, while incorrect assumptions can lead to inaccurate or misleading outcomes. They are essential for reliable engineering analysis.